

CT QC Copilot: Clinical Implementation of an Automated Quality-Control Decision-Support System

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Abstract

We describe the clinical implementation of the CT QC Copilot, an automated decision-support system for computed tomography quality control built on the Physics World Model (PWM) platform. The Copilot automates the end-to-end QC workflow—from DICOM phantom-image ingestion through nine ACR-aligned metric computations, four-layer threshold evaluation, statistical drift detection via Shewhart control charts with Western Electric rules, scored root-cause diagnosis, and triple-output report generation—while preserving the qualified medical physicist’s role as the final decision-maker. In the copilot model, the system computes, the physicist decides: the Copilot drafts the QC report, identifies failing metrics, ranks root-cause hypotheses with confidence scores, recommends the next diagnostic test, and generates trend visualizations, but the physicist retains clinical judgment, accept/reject authority, verification responsibility, and regulatory sign-off. We present the clinical workflow, report structure, drift detection methodology, diagnostic pipeline, and discuss operational considerations for deployment at sites managing 10–50 CT scanners. The system reduces per-scanner QC analysis time from hours of manual work to minutes of automated computation followed by physicist review, enabling trending-based QC as recommended by AAPM Task Group 233. Source code is available at https://github.com/integritynoble/Physics_World_Model.

Keywords: CT quality control, decision support, copilot, statistical process control, Western Electric rules, ACR phantom, AAPM TG-233, clinical implementation

1 Introduction

Quality control of computed tomography scanners is a foundational responsibility of the qualified medical physicist (QMP). The ACR CT Accreditation Program [1] mandates periodic phantom-based testing across multiple performance categories, and AAPM Task Group 233 [2] recommends a shift from simple pass/fail testing to trending-based QC with statistical process control. Yet at most clinical sites, the QC workflow remains manual: a technologist acquires phantom images, the physicist opens vendor-specific analysis software, manually places regions of interest (ROIs), records

values in a spreadsheet, compares against locally maintained thresholds, and writes a summary report.

This manual process faces three scaling challenges. First, a single QMP may oversee 10–50 CT scanners across multiple sites, each requiring monthly QC with annual comprehensive evaluations [2]. At this scale, manual analysis consumes hours per scanner per month. Second, trending-based QC requires maintaining time-series databases and computing control-chart statistics—tasks that are error-prone when performed manually in spreadsheets. Third, when metrics fail, root-cause identification requires correlating multiple artifact signatures against known failure modes—a cognitive task that benefits from systematic computational support.

We present the CT QC Copilot, a decision-support system that automates the computational aspects of CT QC while preserving the physicist’s role as the decision-maker. The term “copilot” is deliberate: the system assists but does not replace the physicist. The copilot model is defined by a clear division of responsibility:

- **The Copilot provides:** automated metric computation, threshold evaluation, drift detection with control charts, scored root-cause hypotheses, “what test next?” recommendations, trend visualizations, and drafted reports.
- **The physicist provides:** clinical judgment, accept/reject/investigate decisions, verification of computational results, regulatory sign-off, and ultimate responsibility for the QC determination.

This division ensures that automation enhances rather than replaces professional judgment, consistent with the role of the QMP as defined by ACR and AAPM.

2 Clinical Workflow

?? illustrates the end-to-end clinical workflow. The Copilot operates in a metric-first mode: all quality metrics are computed directly from reconstructed phantom images and DICOM metadata, with no forward model required for routine QC.

2.1 Step 1: DICOM Ingestion

The technologist acquires ACR CT 464 phantom images using the site’s standard QC acquisition protocol. DICOM files are transferred to the Copilot’s input directory.

The ingestion module performs three safety checks before analysis:

1. **PHI validation:** Twenty PHI-sensitive DICOM tags (PatientName, PatientID, PatientBirthDate, *etc.*) are checked. Seven phantom-pattern regular expressions determine whether the study is a phantom scan. In strict mode (default), non-phantom studies are rejected.

Table 1: Nine QA metrics computed by the CT QC Copilot, with ACR criteria and clinical significance.

#	Metric	ACR Criterion	Clinical Significance
1	CT Number (Water)	0 ± 5 HU	HU calibration accuracy
2	CT Number (Inserts)	Material-specific	Contrast quantification
3	Geometric Accuracy	± 2 mm	Measurement reliability
4	Slice Thickness	± 1.5 mm	Z-axis resolution
5	Uniformity	≤ 5 HU	Field flatness
6	Noise (Std Dev)	vs. baseline	Dose/image quality
7	Low-Contrast Detectability	≥ 4 targets	Soft-tissue visibility
8	Artifact Evaluation	0–3 score	Image integrity
9	Spatial Resolution	≥ 5 lp/cm	Detail visibility

2. **Series selection:** CasePack-defined rules select the target series from potentially multiple series in the study. Selection criteria include series description keywords, slice thickness range, modality, and minimum image count. The first matching rule wins; the selection is logged for audit.

3. **HU rescaling:** Raw pixel values are converted to Hounsfield Units via the DICOM RescaleSlope and RescaleIntercept fields.

The output is a `CTScanBundle`—a vendor-neutral representation containing the 3-D volume in HU, pixel spacing, slice thickness, protocol parameters, and the complete series selection audit log.

2.2 Step 2: Metric Computation

The Copilot computes nine ACR-aligned QA metrics from the ingested volume (Table 1). Phantom center is auto-detected via thresholding and centroid computation. Phantom rotation is estimated from the bone-equivalent insert position to correct ROI placement angles.

Each metric returns a structured result containing the measured value, unit, a derivation record (formula, intermediate values, ACR criterion), and ROI spatial coordinates. Computation of all nine metrics completes in seconds.

2.3 Step 3: Threshold Evaluation

Each metric is evaluated against a four-layer threshold hierarchy:

1. **Standard default:** ACR/AAPM tolerances.
2. **Scanner model:** Manufacturer-specific overrides.
3. **Protocol:** Acquisition-specific adjustments.
4. **Site override:** Institutional customizations.

The most specific applicable threshold is used. Each metric receives a PASS, WARNING, or FAIL status. The overall QC determination is PASS if and only if every metric passes; a single FAIL produces an overall FAIL.

2.4 Step 4: Baseline Comparison

The Copilot compares current measurements against the scanner’s active CommissioningBundle—an immutable, SHA-256-signed baseline snapshot created at commissioning. Per-metric deltas are classified as STABLE ($< 5\%$), DRIFTED ($5\text{--}15\%$), or ALERT ($\geq 15\%$).

2.5 Step 5: Drift Detection

The Copilot applies statistical process control to the scanner’s QC time series. For each metric, a Shewhart control chart is built with center line anchored to the commissioning baseline (when available) and control limits at $\pm 2\sigma$ (warning) and $\pm 3\sigma$ (control).

Five Western Electric rules detect drift patterns:

1. Single point outside $3\sigma \Rightarrow$ FAIL.
2. 2 of 3 consecutive outside 2σ (same side) $\Rightarrow$ ACTION_REQUIRED.
3. 4 of 5 consecutive outside 1σ (same side) $\Rightarrow$ WARNING.
4. Run of 7 on same side of mean $\Rightarrow$ WARNING.
5. Trend of 7 monotone $\Rightarrow$ WARNING.

These rules detect both acute failures (Rule 1) and gradual drift (Rules 3–5) that would be invisible in a single-timepoint pass/fail evaluation. This is the core value proposition of trending-based QC: catching scanner degradation *before* it crosses the action threshold.

2.6 Step 6: Root-Cause Diagnosis (Failure Path)

When one or more metrics fail, the Copilot activates the diagnostic pipeline:

a) Artifact feature extraction. Six quantitative artifact signatures are computed: ring index, cupping index, streak index, HU drift index, noise ratio, and geometric distortion index. Each captures a distinct physical degradation mode.

b) Scored hypothesis ranking. A mismatch library (`clinical_ct_mismatch.yaml`) defines known failure modes with feature weights and expected directions. Each hypothesis is scored as:

$$\text{score}(h) = \underbrace{\sum_{f \in \text{match}} |v_f|w_f}_{\text{evidence}} - \underbrace{\sum_{f \in \text{contradict}} |v_f|w_f}_{\text{penalty}} + \underbrace{0.5 \cdot |\text{matching failed metrics}|}_{\text{bonus}} \quad (1)$$

The Copilot presents the physicist with:

- Ranked hypotheses with confidence levels (high/moderate/low).
- The specific artifact features that support or contradict each hypothesis.
- A recommended next diagnostic test when the top candidates are ambiguous (within 20% score separation).

This structured presentation transforms root-cause identification from an unguided cognitive task into a systematic review of evidence-based hypotheses.

2.7 Step 7: Report Generation

The Copilot produces three output artifacts:

a) JSON report. The `physicist_report.json` is the canonical machine-readable record, containing all metric values, thresholds, baseline deltas, drift alerts, diagnosis results, and a SHA-256 integrity hash.

b) PDF report. The `physicist_report.pdf` is formatted for review and sign-off:

- Header with scanner ID, date, CasePack version.
- Color-coded PASS/FAIL summary box.
- Eight-column metrics table (metric, value, unit, baseline, standard threshold, applied threshold, layer, status).
- Drift alerts section with Western Electric rule details.
- Diagnosis section with ranked hypotheses (failure path only).
- Signature block for physicist sign-off.
- Compliance footer referencing ACR and AAPM standards.

c) Evidence folder. The `evidence/` directory contains ROI overlay images showing where measurements were taken, trend plots for each trended metric, and per-metric derivation logs in JSON.

2.8 Step 8: Physicist Review and Sign-Off

The physicist reviews the Copilot's output:

1. Verifies the overall PASS/FAIL determination.
2. Reviews any WARNING or FAIL metrics against the evidence artifacts.

3. For failures, evaluates the ranked hypotheses and decides whether to accept the top hypothesis, investigate further, or request a rescan.
4. Signs the PDF report, which constitutes the official QC record.

The physicist may override any Copilot determination. The Copilot’s role ends with recommendation; the physicist’s judgment is authoritative.

3 Operational Considerations

3.1 Scaling to Multi-Scanner Sites

A physicist overseeing N scanners with monthly QC faces $12N$ analysis sessions per year. At $N = 30$ (a typical large health system), this is 360 sessions. With manual analysis at 1–2 hours per session, the annual burden is 360–720 hours of QC analysis alone.

The Copilot reduces the per-session time from hours to minutes: automated computation completes in seconds, and physicist review of a PASS result requires only verification of the summary and spot-checking of a few metrics. FAIL results require more attention but benefit from the structured diagnostic output.

3.2 Trending-Based QC

AAPM TG-233 advocates replacing binary pass/fail testing with trending-based QC [2]. The Copilot implements this recommendation directly: every measurement is added to the scanner’s time series, control charts are updated automatically, and drift alerts are generated based on Western Electric rules.

The practical benefit is early warning. A gradual increase in noise standard deviation, for example, may not cross the ACR pass/fail threshold for months, but a Western Electric Rule 5 (monotone trend of 7 points) would flag it early, giving the site time to schedule preventive maintenance before the scanner fails accreditation criteria.

3.3 Commissioning and Service Events

The Copilot’s baseline system handles the full lifecycle:

- **Initial commissioning:** A new `CommissioningBundle` is created with all metrics, acquisition parameters, and physicist approval.
- **Service events** (tube change, software upgrade): A new baseline version is created and chained to the previous one. The drift detector resets its center line to the new baseline while preserving the historical time series for review.
- **Annual QC:** The current measurements are compared against the active baseline and the full time series.

The version chain ensures that the complete commissioning history of a scanner is always available, from installation through every service event to the present.

3.4 Regulatory Scope

The CT QC Copilot is a decision-support tool, not a standalone diagnostic system. In its current version (v1.0), it is intended for research and internal QA use. The system does not make clinical decisions, does not process patient data (phantom-based only), and does not replace the qualified medical physicist’s judgment.

Deployment as a regulated Software as a Medical Device (SaMD) would require additional validation, 510(k) or De Novo submission (depending on classification), and compliance with IEC 62304 software lifecycle requirements. The platform’s extensive audit trails, version-controlled CasePacks, and SHA-256 integrity hashing provide a foundation for future regulatory submissions.

4 Integration with the Physics World Model

The CT QC Copilot is one application within the Physics World Model (PWM) framework. PWM models physical measurement as a pipeline of operator-graph transformations, where each transformation can introduce mismatch between the assumed and actual physics.

In the CT QC context, the dominant mismatch sources map to the Triad Law’s three gates:

- **Gate 1 (Protocol design):** Inadequate scan parameters—insufficient angular sampling, coverage gaps, wrong reconstruction kernel.
- **Gate 2 (Dose budget):** Sub-optimal mAs or kVp selection leading to elevated noise or contrast degradation.
- **Gate 3 (Scanner calibration):** Drift in HU calibration, detector gain, center-of-rotation, or geometric alignment—the dominant failure mode in routine clinical QC.

The Copilot’s nine QA metrics directly probe Gate 3 degradation modes. When degradation is detected, the diagnosis engine identifies which specific calibration parameter has drifted, and the optional operator-graph module (`troubleshoot_mode=True`) can quantify the mismatch magnitude via reprojection error analysis.

5 Discussion

The CT QC Copilot addresses a practical clinical need: automated, reproducible, trending-based CT quality control with structured decision support. Three aspects merit discussion.

The copilot model. The decision not to fully automate QC—*i.e.*, to keep the physicist in the loop—is deliberate. Full automation would risk de-skilling the physicist, obscuring failure modes that require clinical context, and creating regulatory complications. The copilot model preserves professional expertise while eliminating the mechanical burden of manual computation and data entry.

Comparison with existing tools. Commercial CT QC software (vendor-provided analysis packages, third-party tools) typically operates as a black box: the user places ROIs, the software computes metrics, and the results are displayed. The Copilot differs in three respects: (i) the analysis logic is fully transparent via CasePacks; (ii) drift detection and diagnosis are integrated rather than bolt-on; and (iii) the triple-output report provides machine-readable, human-readable, and evidentiary records simultaneously.

Limitations. The current implementation has been designed for the ACR CT 464 phantom. Extension to other phantom types (Catphan, vendor-specific phantoms) requires new CasePack definitions and validation. The diagnosis engine’s mismatch library is expert-curated; a data-driven approach could improve coverage as multi-site QC data accumulates. The system has not been prospectively validated in a multi-site clinical trial; such validation is an important next step.

Future directions. Planned extensions include: (i) CasePacks for CBCT and adaptive radiotherapy QA, expanding the Copilot’s utility in the therapeutic physics domain; (ii) integration with PACS and RIS for automated QC scheduling and result archival; (iii) multi-site data aggregation for data-driven mismatch library enrichment; and (iv) a web-based dashboard for fleet-wide scanner status monitoring.

6 Conclusion

The CT QC Copilot provides automated, reproducible, trending-based CT quality control as a decision-support system for the qualified medical physicist. By automating metric computation, threshold evaluation, drift detection, and root-cause diagnosis while preserving the physicist’s decision authority, the system enables a single physicist to efficiently oversee large scanner fleets with consistent, evidence-based QC. The copilot model—the system computes, the physicist decides—offers a practical path to implementing AAPM TG-233 trending-based QC recommendations in clinical practice.

Data and code availability. All source code is available at https://github.com/integritynoble/Physics_World_Model and <https://solveeverything.org>. No non-public datasets were used. All analyses are phantom-based; no patient data were used.

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